

Pre-Trained Models – Complete Reference with website link

Category	Purpose	Model Name	Documentation / Official Link
Image Classification	Identify objects in images	VGG16	VGG16 Documentation
Image Classification	Identify objects in images	VGG19	VGG19 Documentation
Image Classification	Identify objects in images	ResNet50	ResNet50 Documentation
Image Classification	Identify objects in images	InceptionV3	InceptionV3 Documentation
Image Classification	Identify objects in images	Xception	Xception Documentation
Image Classification	Identify objects in images	DenseNet121	DenseNet Documentation
Image Classification	Identify objects in images	EfficientNetB0	EfficientNet Documentation
Image Classification	Identify objects in images	MobileNetV2	MobileNet Documentation
Object Detection	Detect and locate objects	YOLO	YOLO Documentation
Object Detection	Detect and locate objects	YOLOv8	YOLOv8 Documentation
Object Detection	Detect and locate objects	Faster R-CNN	TensorFlow Object Detection API
Object Detection	Detect and locate objects	Mask R-CNN	Mask R-CNN GitHub
Object Detection	Detect and locate objects	SSD	SSD Documentation
Object Detection	Detect and locate objects	RetinaNet	RetinaNet Documentation
Face Recognition	Identify and verify faces	FaceNet	FaceNet GitHub
Face Recognition	Identify and verify faces	DeepFace	DeepFace GitHub

Face Recognition	Identify and verify faces	ArcFace	ArcFace GitHub
Face Recognition	Identify and verify faces	VGG-Face	VGG-Face Project
Face Recognition	Identify and verify faces	OpenFace	OpenFace Project
Image Segmentation	Identify exact object boundaries	Mask R-CNN	Mask R-CNN GitHub
Image Segmentation	Identify exact object boundaries	DeepLabV3	DeepLab Documentation
Image Segmentation	Identify exact object boundaries	U-Net	U-Net Project
Text / NLP	Understand text	BERT	BERT Documentation
Text / NLP	Generate text	GPT	OpenAI Platform
Text / NLP	Text-to-Text Processing	T5	T5 Documentation
Text / NLP	Language Understanding	RoBERTa	RoBERTa Documentation
Text / NLP	Language Modeling	XLNet	XLNet Documentation
Text / NLP	Lightweight NLP	DistilBERT	DistilBERT Documentation
Speech Recognition	Convert speech to text	Whisper	Whisper GitHub
Speech Recognition	Convert speech to text	DeepSpeech	DeepSpeech GitHub
Image Generation	Generate images from text	DALL·E	DALL·E Information
Image Generation	Generate realistic images	GAN	GAN Paper
Image Generation	Generate human faces	StyleGAN	StyleGAN Project

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N o	Category	Model	Purpose	Real-Time Example
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1	Image Classification	AlexNet	Image Classification	Cat vs Dog Recognition
2	Image Classification	LeNet-5	Digit Recognition	Bank Cheque Reading
3	Image Classification	GoogLeNet	Image Recognition	Google Photos
4	Image Classification	NASNet	Auto Architecture Search	Automated AI Systems
5	Image Classification	ShuffleNet	Mobile Vision	Smartphone AI Camera
6	Image Classification	SqueezeNet	Lightweight Classification	IoT Cameras
7	Image Classification	ConvNeXt	Modern Image Recognition	Medical Imaging
8	Image Classification	Vision Transformer (ViT)	Image Classification	Satellite Analysis
9	Image Classification	Swin Transformer	Computer Vision	Autonomous Vehicles
10	Image Classification	RegNet	Efficient Classification	Industrial Inspection
11	Object Detection	YOLOv3	Real-Time Detection	CCTV Monitoring
12	Object Detection	YOLOv4	Fast Detection	Smart Cities
13	Object Detection	YOLOv5	Industrial Detection	Factory Monitoring
14	Object Detection	YOLOv6	Edge AI Detection	Smart Cameras
15	Object Detection	YOLOv7	High-Speed Detection	Traffic Analytics
16	Object Detection	DETR	Transformer Detection	Autonomous Driving
17	Object Detection	EfficientDet	Efficient Detection	Drone Monitoring
18	Object Detection	CenterNet	Object Tracking	Retail Analytics
19	Object Detection	Cascade R-CNN	High Accuracy Detection	Security Systems

20	Object Detection	R-FCN	Region-Based Detection	Warehouse Automation
21	Face Recognition	SphereFace	Face Verification	Security Gates
22	Face Recognition	CosFace	Face Authentication	Employee Attendance
23	Face Recognition	InsightFace	Face Analysis	Airport Security
24	Face Recognition	RetinaFace	Face Detection	Smartphone Cameras
25	Face Recognition	Dlib Face Recognition	Face Matching	Visitor Management
26	Image Segmentation	SegNet	Semantic Segmentation	Road Detection
27	Image Segmentation	FCN	Pixel Classification	Medical Imaging
28	Image Segmentation	PSPNet	Scene Parsing	Autonomous Driving
29	Image Segmentation	HRNet	Human Pose Segmentation	Fitness Apps
30	Image Segmentation	BiSeNet	Real-Time Segmentation	Face Parsing
31	NLP	ALBERT	Efficient Language Understanding	Chatbots
32	NLP	ELECTRA	Text Classification	Spam Detection
33	NLP	DeBERTa	Language Understanding	Search Engines
34	NLP	BART	Text Summarization	News Summaries
35	NLP	Pegasus	Document Summarization	Research Papers
36	NLP	LLaMA	Large Language Model	AI Assistants
37	NLP	Mistral	Text Generation	Business Chatbots

38	NLP	Falcon	Conversational AI	Customer Support
39	NLP	Gemma	Language Processing	AI Applications
40	NLP	Claude Model Family	Text Understanding	AI Assistants
41	Speech Recognition	Wav2Vec 2.0	Speech Recognition	Voice Assistants
42	Speech Recognition	HuBERT	Speech Processing	Call Centers
43	Speech Recognition	Speech2Text	Speech-to-Text	Meeting Transcription
44	Speech Recognition	Jasper	Voice Recognition	Virtual Assistants
45	Speech Recognition	QuartzNet	Audio Processing	Smart Speakers
46	Image Generation	Stable Diffusion	Text-to-Image	AI Art Creation
47	Image Generation	Midjourney Model	AI Image Generation	Creative Design
48	Image Generation	CycleGAN	Image Translation	Day-to-Night Conversion
49	Image Generation	Pix2Pix	Image Transformation	Sketch-to-Photo
50	Image Generation	DCGAN	Synthetic Image Creation	Product Visualization

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Image Classification → 18+ Models
Object Detection → 16+ Models
Face Recognition → 10+ Models
Image Segmentation → 8+ Models
Text / NLP → 15+ Models
Speech Recognition → 7+ Models
Image Generation → 10+ Models

